Data Mining Assignment 1 Report

Frequent Pattern Mining

March 03, 2026

Ilias Sarikakis
20240433

The source code of this report can be found in the following [GitHub repository].

1 Introduction

The goal of this assignment is to apply the popular frequent pattern mining algorithm apriori on a dataset containing information related to the transactional behavior of an online grocery retailer's customers, uncover association rules and useful marketing insights.

2 Data Inspection and Preparation (Task 1)

2.1 Data Understanding

The dataset used in this exercise is split into 5 files, each of which contains different information related to the transactional information of the online grocery retailer. Specifically the files contain the following information:

1. **aisles.csv** contains 134 unique aisles alongside the name associated with each aisle.
2. **departments.csv** contains 21 unique departments alongside the name associated with each department.
3. **orders.csv** is the main file containing the order history of all users in the dataset. Specifically it includes 3,421,083 unique orders associated with 206,209 unique users. All users have between 4 and 100 orders, with the average being ≈17 (16.59) orders per user. Additionally, the file contains information related to the moment when each order was placed, such as the day of the week and the hour of the day. Finally the file also keeps track of how many days have passed since the previous order of each user.
4. **products.csv** contains 49,688 unique products, alongside the name associated with each product and the aisle and department corresponding to each product.
5. **order_products.csv** contains all the products that are part of each order. Specifically there are a total of 3,346,083 unique order IDs (75,000 less than the orders found in orders.csv) with on average ≈10 (10.1) products, ranging from 1 to 145 products per order. Additionally, this file keeps track of the order each product was added to the cart during the order creation, as well as if certain products have been reordered by each user in the past.

Figure 1 contains a diagram depicting how each of the 5 files of the dataset are connected with each other. Each product is associated with a single aisle and department ID, while at the same time *Order_Products* is connected with the *Orders* and *Products* data in a one-to-many association.

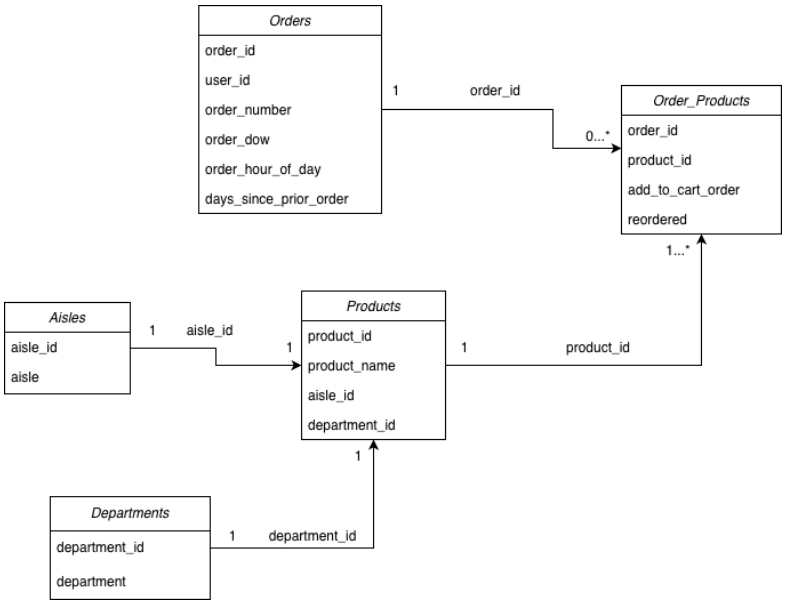

Figure 1: Dataset associations diagram

2.2 Data Pruning

Due to the dataset's ≈3.4 million orders making mining association rules a trivial task computationally, the decision was taken to reduce the number of considered orders.

2.2.1 Pruning Proposal

First, the basket size of every order was computed as the number of unique products appearing in that order. Inspecting the resulting distribution depicted in Figure 2 highlighted that the vast majority of the orders contained less than 20 products, with over 150,000 orders even containing 1 product.

Since single-item orders cannot contribute to association rules with a minimum length of two items, all orders consisting of 1 product were removed.

On the remaining orders, proportional stratified random sampling was applied. Specifically, the orders were stratified into buckets based on the number of products they contained. Specifically the orders were broken down into 4 buckets containing 2-4, 5-9, 10-14, and 15+ total products respectively. This stratification ensures that the final sample retains a realistic mix of order sizes rather than being dominated by the most common basket sizes.

Figure 2: Distribution of unique number of products per order on all ≈3.4 million orders

The proportion of each bucket was calculated on the whole dataset and then it was translated into a target number of sampled orders. In total 200,000 orders were sampled from the original dataset based on the calculated proportions. To ensure that the taken sample did not exhibit any unusual behavior in terms of the total number of products per order, the distribution was recalculated on the sampled dataset, resulting in the plot depicted in Figure 3.

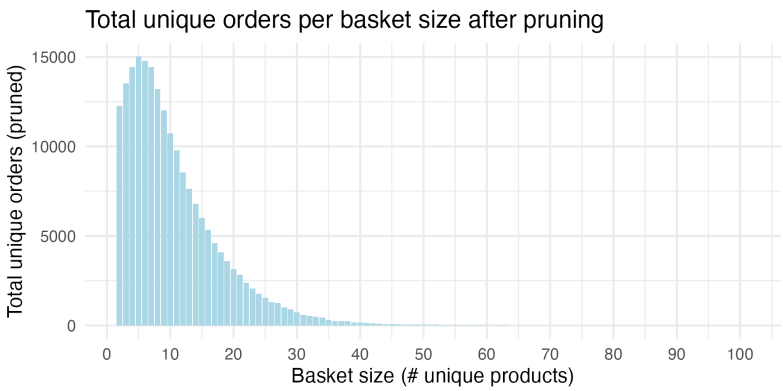

Figure 3: Distribution of unique number of products per order after applying proportional stratified random sampling

2.2.2 Alternative Pruning Options

Another alternative pruning option that was considered was to sample the top-k number of most frequent products per aisle. Afterwards only the orders containing these sampled products would remain, effectively reducing the number of products per order.

This approach would however inflict bias towards the most popular products, which could effectively remove niche, yet still actual associations. Thus a decision was taken to proceed with the fairer proportion stratified random sampling approach.

2.3 Constructing Transaction

Before mining for association rules using the Apriori algorithm, the data need to be transformed into transactions. The 200,000 pruned orders are originally in a long format, where an *order_id* contains multiple entries, each with a different *product_id*.

For the purposes of this assignment, the goal is to identify distinct rules on (1) the overall categories of the products in each order, described as aisles, and (2) on the timing of when orders were placed, alongside which product categories were purchased.

To achieve this, first the metadata of the aisles and the time of day when the orders were placed were collected and joined with the pruned orders, forming the table shown in Figure 4.

	order_id	product_id	add_to_cart_order	reordered	order_hour_of_day	aisle_id	aisle
1	25	9755	1	1	10	23	popcorn jerky
2	25	31487	2	0	10	23	popcorn jerky
3	25	37510	3	1	10	116	frozen produce
4	25	14576	4	1	10	52	frozen breakfast
5	25	22105	5	0	10	21	packaged cheese
6	25	1934	6	1	10	21	packaged cheese
7	25	6383	7	1	10	86	eggs
8	25	39046	8	0	10	106	hot dogs bacon sausage
9	25	48171	9	0	10	129	frozen appetizers sides
10	25	33506	10	1	10	40	dog food care
11	25	30320	11	1	10	79	frozen pizza
12	25	15925	12	0	10	38	frozen meals
13	25	37449	13	0	10	21	packaged cheese
14	25	48299	14	1	10	123	packaged vegetables fruits
15	29	43352	1	1	12	32	packaged produce
16	29	21386	2	1	12	115	water seltzer sparkling water
17	29	13176	3	1	12	24	fresh fruits
18	29	8571	4	1	12	32	packaged produce
19	29	33768	5	1	12	83	fresh vegetables

Figure 4: Pruned orders table containing all relevant metadata

Afterwards, the time of day was converted to time periods during the day. Specifically orders placed between 6 and 11 were considered "morning" orders, 12 to 17 "afternoon" orders, 18 to 22 "evening" orders and the remaining as "night" orders.

The final step to create the transactions is to convert the orders into a list where each row represents an order and the values contain the time period associated with each order, as well as all relevant product aisles for the respective order, resulting in a list similar to Figure 5. For example, The second order was placed during the afternoon and it contained packaged produce, sparkling water, fresh fruits and fresh vegetables.

```
> head(timing_transactions_list)
[[1]]
 [1] "TOD_morning"                "popcorn jerky"              "frozen produce"             "frozen breakfast"           "packaged cheese"
 [6] "eggs"                       "hot dogs bacon sausage"     "frozen appetizers sides"    "dog food care"              "frozen pizza"
[11] "frozen meals"               "packaged vegetables fruits"

[[2]]
[1] "TOD_afternoon"                 "packaged produce"              "water seltzer sparkling water" "fresh fruits"
[5] "fresh vegetables"

[[3]]
[1] "TOD_afternoon"              "doughs gelatins bake mixes" "fresh fruits"               "frozen produce"             "cookies cakes"
[6] "frozen dessert"             "other creams cheeses"
```

Figure 5: Transactions list containing the time period and all relevant aisles of each order

Finally, the transactions list is converted to a `transactions` object from the **arules R package**, a binary incidence matrix which coerces the list to produce a transactions data set with the names of the items in the list being used as item labels.

3 Mining Association Rules (Task 2)

3.1 Association Rules

Running the **apriori** function from **arules** using the sampled 200,000 transactions with a minimum of 0.01 support, 0.01 confidence, minimum length of 2 and maximum length of 3 resulted in the identification of 7,881 association rules as it can be observed in Figure 6.

```
> rules_timing <- apriori(
+   transactions_timing,
+   parameter = list(supp = 0.01, conf = 0.01, minlen = 2, maxlen = 3)
+ )
Apriori

Parameter specification:
 confidence minval smax arem  aval originalSupport maxtime support minlen maxlen target  ext
       0.01    0.1    1 none FALSE            TRUE       5    0.01      2      3  rules TRUE

Algorithmic control:
 filter tree heap memopt load sort verbose
    0.1 TRUE TRUE  FALSE TRUE    2    TRUE

Absolute minimum support count: 2000

set item appearances ...[0 item(s)] done [0.00s].
set transactions ...[138 item(s), 200000 transaction(s)] done [0.06s].
sorting and recoding items ... [102 item(s)] done [0.01s].
creating transaction tree ... done [0.05s].
checking subsets of size 1 2 3 done [0.06s].
writing ... [7881 rule(s)] done [0.00s].
creating S4 object  ... done [0.01s].
```

Figure 6: Running apriori on the 200,000 orders with support 0.01 and confidence 0.01

Running apriori with varying support and confidence value combinations with a minimum length of 2 and a maximum of 3 results in the table depicted in Figure 7. It is observed that support and confidence act as filtering thresholds that control which patterns and rules are considered meaningful.

A high support value reduces the number of frequent itemsets by eliminating rare combinations. Decreasing support allows for more rare combinations to be considered, thus increasing the number of association rules that can be derived.

Confidence measures how often one item is purchased together with another, given that the first item has already been purchased. In other words, it indicates the reliability of a rule by showing the percentage of transactions containing the first item that also contain the second item. A higher confidence value means the co-occurrence is more consistent and therefore the rule is stronger.

	support	confidence	n_rules
1	0.100	0.100	81
2	0.100	0.010	81
3	0.100	0.001	81
4	0.010	0.100	6895
5	0.010	0.010	7881
6	0.010	0.001	7881
7	0.001	0.100	90201
8	0.001	0.010	120712
9	0.001	0.001	121374

Figure 7: Identified associations of the 200,000 sampled orders with varying support and confidence parameters with minimum length of 2 and maximum length of 3

Figure 7 clearly illustrates the dominant effect of the support threshold on the number of generated rules. Reducing support from 0.1 to 0.001 increases the number of rules from tens to over a hundred thousand, demonstrating the combinatorial growth of frequent itemsets. In contrast, varying confidence while keeping support fixed results in comparatively minor changes in rule count. This indicates that support primarily controls the size and computational complexity of the rule set, whereas confidence mainly filters the strength of the discovered associations.

3.2 Market Insights

Looking at the generated association rules, the following product category insights can be derived:

1. **Fresh fruits and fresh vegetables**. In Figure 8 it can be observed that there exists a strong, symmetric association between fresh vegetables and fresh fruits, with support of 33.4%. Interestingly, customers who buy fresh vegetables purchase fresh fruits with a much higher confidence (72.1%) compared to customers purchasing fresh fruits also purchasing fresh vegetables (57.4%). Despite the fresh fruits being purchased more broadly (58.1% vs 46.3%), fresh vegetables appear to be a much stronger predictor. From a retail perspective, this supports cross-promotions or bundling, particularly targeting fresh vegetable buyers to increase fresh fruit sales.

rules	support	confidence	coverage	lift	count
{fresh vegetables} => {fresh fruits}	0.334095	0.7211047	0.463310	1.2406102	66819
{fresh fruits} => {fresh vegetables}	0.334095	0.5747871	0.581250	1.2406102	66819

Figure 8: Association of fresh fruits and fresh vegetables

2. **Granola and yogurt**. In Figure 9 it can be observed that there exists a meaningful positive association between granola and yogurt. Although support for this rule is low at 1.61%, the confidence is fairly strong at 54%. This means that more than half of granola buyers also purchase yogurt. The observed lift of 1.97 indicates that this co-occurrence is nearly twice as likely as expected under independence, which makes sense considering that granola and yogurt are a popular breakfast option. This is a practical cross-selling opportunity, where promotions for these two items could be justified to attract more customers.

rules	support	confidence	coverage	lift	count
{granola} => {	All	All	All	All	All
{granola} => {yogurt}	0.01614	0.5404319	0.029865	1.973208	3228

Figure 9: Association of granola and yogurt

3. **Pasta, packaged cheese and pasta sauce**. In Figure 10, although at first glance it appears that the association of dry pasta, packaged cheese and pasta sauce does not exhibit high support (1%) and the confidence is relatively low at 32.3%, the observed lift of 4.9 indicates that these products are almost 5 times more likely to be bought together than independently. This rule showcases a very niche, yet very strong association which could be leveraged from a business perspective with bundles or special discounts to incentivize the customers to purchase these items more often.

rules	support	confidence	coverage	lift	count
{dry pasta,packaged cheese} => {pasta sauce}	0.010300	0.32344167	0.031845	4.941436	2060

Figure 10: Association of dry pasta, packaged cheese and pasta sauce

Similarly, the time of day when the orders were placed could be leveraged to uncover some very interesting insights, such as:

1. **Afternoon demand for fresh fruits and fresh vegetables**. From Figure 11 it can be observed that a strong association exists between orders placed during the afternoon hours (12:00 - 17:00) and the inclusion of fresh fruits or fresh vegetables with 57.2% and 47.3% confidence respectively. Despite that, the lift hovers at around 1, indicating that there exists a deman for fresh fruits and vegetables throughout the day and not just during the afternoons. This could be an interesting business insight in order to ensure that there exists a steady supply of fresh produce during the afternoon.

rules	support	confidence	coverage	lift	count
{TOD_afternoon} => {fresh fruits}	0.271670	0.57292589	0.474180	0.9856790	54334
{TOD_afternoon} => {fresh vegetables}	0.224360	0.47315365	0.474180	1.0212464	44872

Figure 11: Association of afternoon with fruits and vegetables

2. **Morning yogurt with breakfast bakery**. A more niche insight could be obtained from Figure 12, where there exists an association between orders placed in the morning containing yogurt with products from the breakfast bakery. Although the support and confidence are low, at 1% and 10.9% respectively, a strong argument can be made about lift of 1.5, which shows that

customers who purchase yogurt in the morning, are also purchasing breakfast from the bakery. This could be an interesting opportunity to provide special promotions or combos during the morning, when customers may consider it for their breakfast option.

rules	support	confidence	coverage	lift	count
TOD_morning	All	All	All	All	All
{TOD_morning,yogurt} => {breakfast bakery}	0.010165	0.10987407	0.092515	1.5168644	2033

Figure 12: Association of morning orders containing yogurt with breakfast bakery

3. **Afternoon tortillas with packaged cheese**. Another interesting association can be observed in Figure 13 with orders placed in the afternoon containing tortillas and packaged cheese. Although the support for this rule is low at 1.3%, its confidence is reasonably high at 44.9%. Furthermore the lift also indicates a strong correlation between tortillas and cheese during the afternoon with a value of 1.86. Overall this suggests a meal-based pattern, likely related to lunch or dinner preparation. From a managerial perspective, this rule supports targeted afternoon promotions, bundle offers, or cross-merchandising strategies in these categories. While not a high-volume pattern, it represents a strong and actionable niche behavior driven by complementary product demand.

rules	support	confidence	coverage	lift	count
{TOD_afternoon,tortillas flat bread} => {packaged cheese	All	All	All	All	All
{TOD_afternoon,tortillas flat bread} => {packaged cheese}	0.013085	0.4498883	0.029085	1.860503	2617

Figure 13: Association of afternoon orders containing tortillas with packaged cheese